

Welcome to the

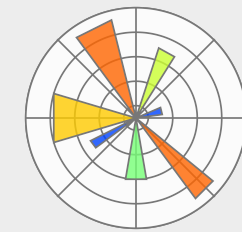
BoBiAC

Boston Bioimage Analysis Course | 2026

 **CITE:** cite.hms.harvard.edu

 **Date:** 6th -11th July 2025, 9:00 am - 6:30 pm

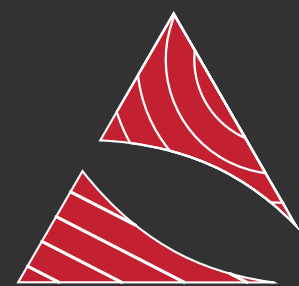
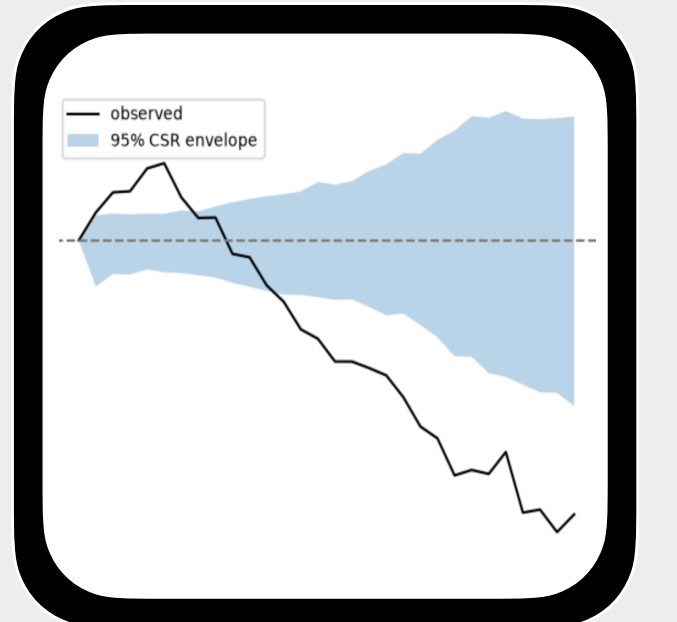
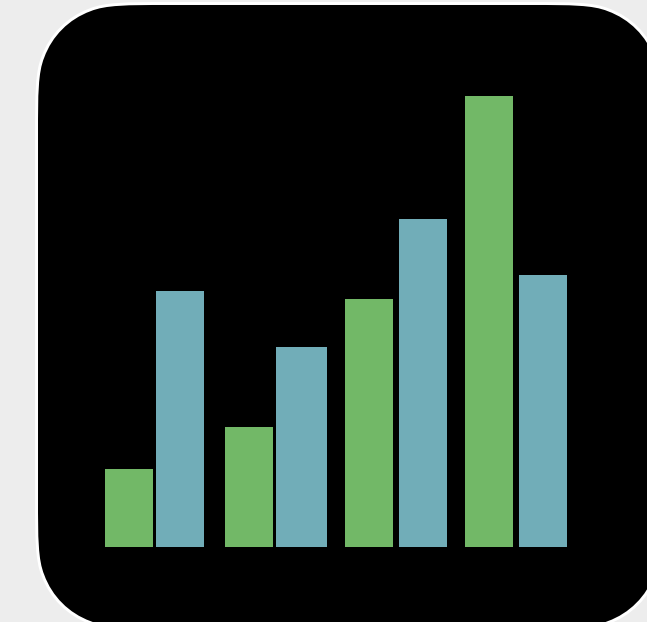
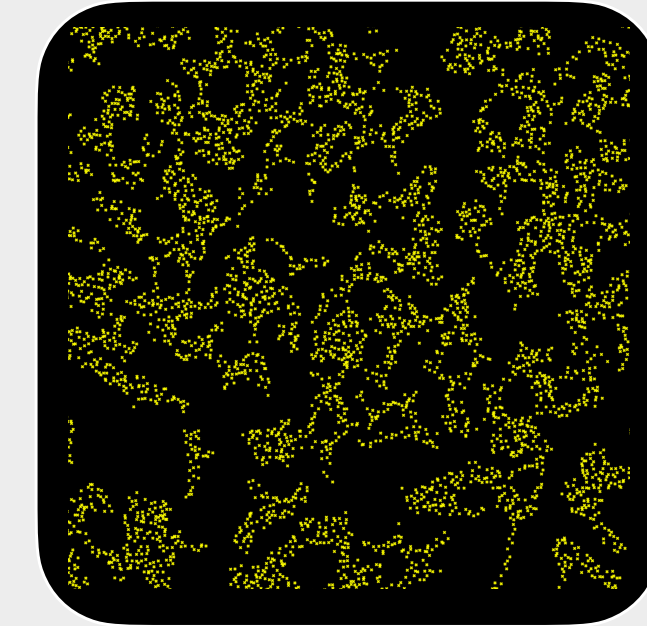
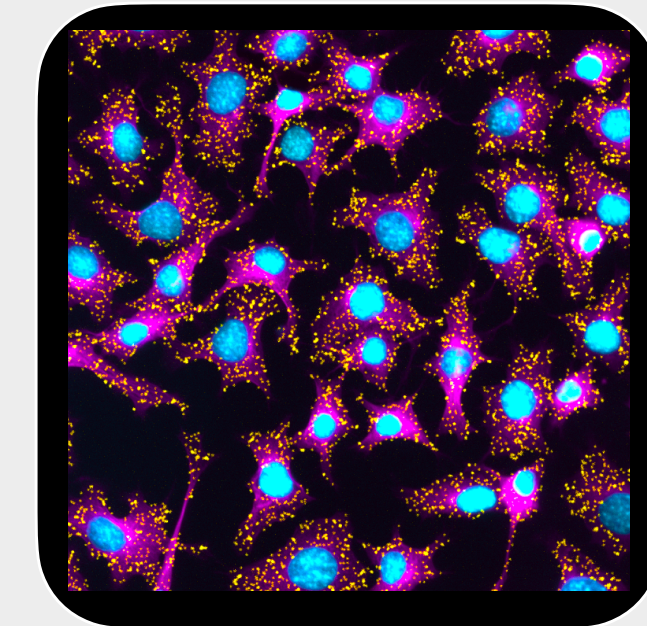
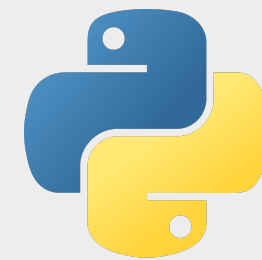
 **Time:** 9:00 am - 6:30 pm



 **location:** Gordon Hall - Harvard Medical School

 **website:** bobiac.github.io

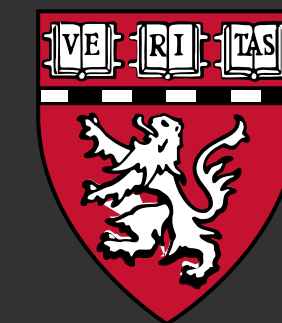
 **course material:** bobiac.github.io/bobiac-book



CITE
Core for Imaging Technology & Education



**BioImaging
North America**



HARVARD
MEDICAL SCHOOL



The BoBiAC Team



CITE: cite.hms.harvard.edu



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Director
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Harvard Medical School



Max Brambach, PhD

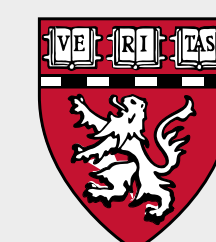
Postdoc
Oyler-Yaniv Lab
Harvard Medical School



Amelie Andreas

Research Assistant
Paulsson Lab
Harvard Medical School

 Organizer & Instructor  Teaching Assistant



General Information

Internal communications:

- Slack channels (mainly #course-announcements & DMs)

location:

- Harvard Medical School - Gordon Hall - Room 106

time:

- 9:00 am - 6.30 pm
- optional office hours from 7:30 pm

breakfast:

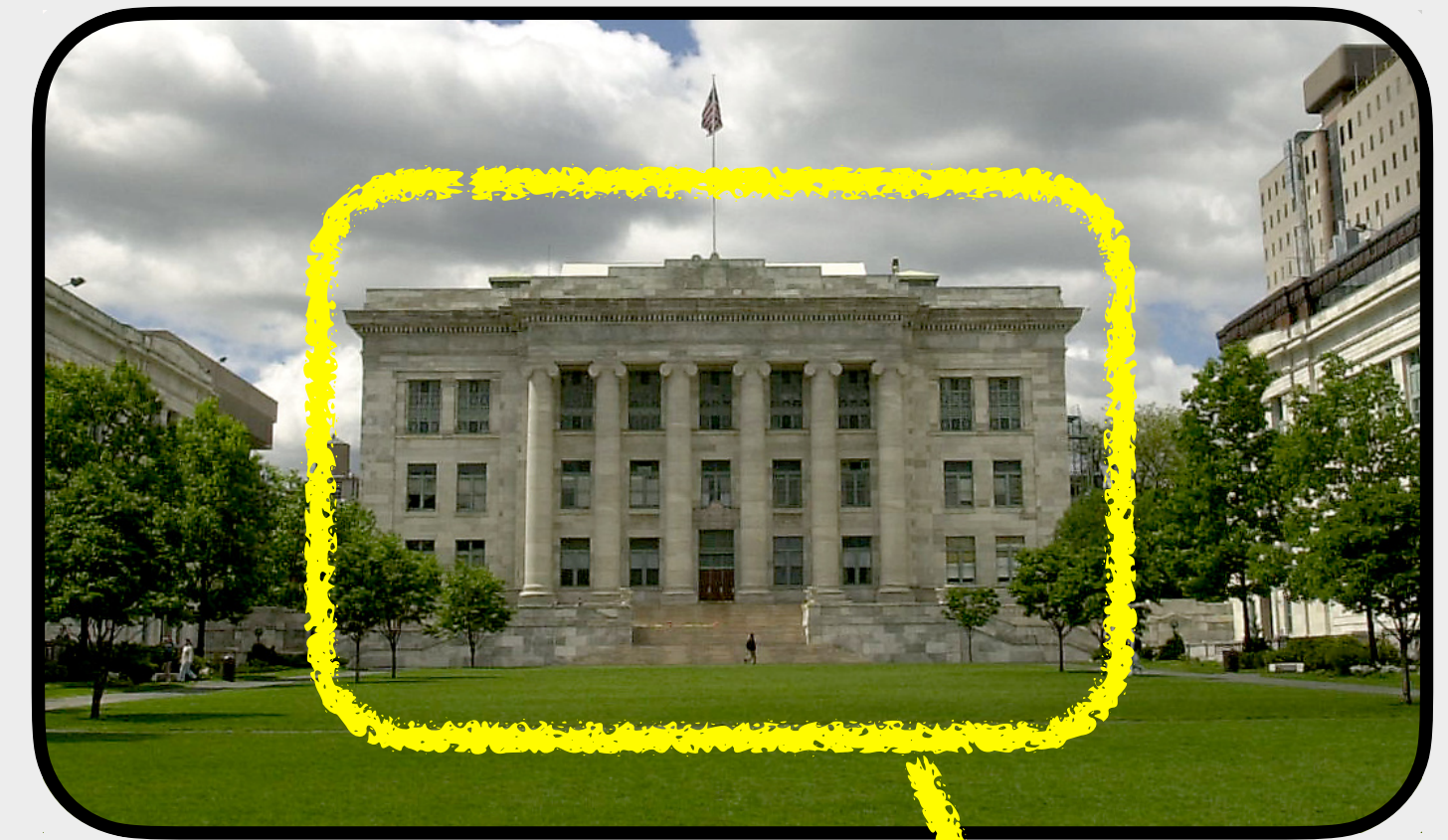
- bagels 🥯, coffee, tea & snacks @ 9:00 am

lunch:

- 12:00 pm in Gordon Hall, Room 106

social dinner:

- Thursday July 9th @ 7:00 pm
- Shy Bird (201 Brookline Ave, Boston - 10 min walk)
- we leave @ 6:30 pm (🚶)





WiFi Connection



- ▶ **Harvard Secure:** if you have Harvard credentials
- ▶ **Eduroam:** if you have academic credential (edu)
- ▶ **Harvard Guest:** you will need to register your laptop for that (if you provided the MAC address in advance, you should be able to directly connect)
- ▶ If all of above do not work, let us know!





- ▶ The course is designed for **beginners** in python and image analysis
- ▶ Each day consists of a **mix of lectures** explaining key bioimage analysis concepts, interspersed with **practical, hands-on exercises** using **Python**.
- ▶ Every morning, we will **start** the day **with coffee** and **open discussion** (Q&A).
- ▶ Most of these **exercises** will be completed either **step-by-step as a class** or in small groups.
- ▶ The course should be **interactive**, there are absolutely **no stupid questions**, you are encouraged to ask questions. During the practical exercise feel free to consult with your neighbors, you can help each other to understand concept.
- ▶ Each day (first four days), five participants will give a **brief 3–5 minute introduction** using their prepared slide (introduce yourself, your work, and the image analysis tasks you hope to perform).





Schedule



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	Monday 6th July	Tuesday 7th July	Wednesday 8th July	Thursday 9th July	Friday 10th July	Saturday 11th July						
9:00am - 9:30am	Welcome + Intro	Coffee & Questions?	Coffee & Questions?	Coffee & Questions?	Coffee & Questions?	Coffee & Questions?						
9:30am - 10:00am	Coffee Break	Error Messages	Segmentation (Machine Learning)	Segmentation (Deep Learning)	Single Cell Measurements	Cell Population Measurements (Spatial Statistics & Colocalization)						
10:00am - 10:30am	Getting Started with Python	Python Bioimage Analysis										
10:30am - 11:00am												
11:00am - 11:15am	The Python Basics	Coffee Break	Coffee Break	Coffee Break	Coffee Break	Coffee Break						
11:15am - 11:30am		Segmentation (Classic)	Segmentation (Deep Learning)	Deep Learning Spot Detection	Single Cell Measurements	Cell Population Measurements (Spatial Statistics & Colocalization)						
11:30am - 12:00pm	Laptops Setup											
12:00pm - 12:30pm	Lunch	Lunch	Lunch	Lunch	Lunch	Lunch						
12:30pm - 1:00pm												
1:00pm - 1:30pm	Student Presentations (5x5min)	Student Presentations (5x5min)	Student Presentations (5x5min)	Student Presentations (5x5min)	Cell Population Measurements (Spatial Statistics & Colocalization)	Student Group Work						
1:30pm - 2:00pm	The Python Basics	Segmentation (Classic)	Segmentation (Deep Learning)	Managing your own Python Workflow								
2:00pm - 2:30pm				Student Group Work		How to Acquire Analyzable Microscopy Images						
2:30pm - 3:00pm	Introduction to Digital Images											
3:00pm - 3:30pm												
3:30pm - 4:00pm	Coffee Break	Coffee Break	Coffee Break	Coffee Break	Coffee Break	Coffee Break						
4:00pm - 4:30pm	Python for Bioimage Analysis	Segmentation (Machine Learning)	Segmentation (Deep Learning)	Student Group Work	Cell Population Measurements (Spatial Statistics & Colocalization)	Using Python in the Real World						
4:30pm - 5:00pm												
5:00pm - 5:30pm				Measurements and Quantification								
5:30pm - 6:00pm	Coffee Break	Coffee Break	Coffee Break	Coffee Break	Coffee Break	Coffee Break						
6:00pm - 6:30pm	Python for Bioimage Analysis	Segmentation (Machine Learning)	Segmentation (Deep Learning)	Measurements and Quantification	Measurements and Quantification	Feedback & Wrap-Up						
6:30pm - 7:30pm				Social Dinner								
7:30pm - 8:30pm	Optional: Office Hour	Optional: Office Hour	Optional: Office Hour		Optional: Office Hour							
8:30pm - 9:30pm												





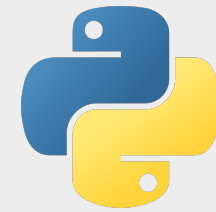
Course Topics I



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Learning Python Basics



- ▶ **Getting Started with Python and uv:** what is python? How do I install it? How do I use it?
- ▶ **The Python Basics:** how do I write python code? What is the syntax?





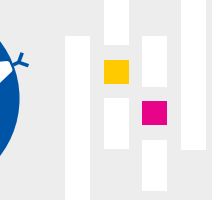
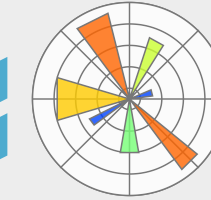
Course Topics II



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Learning Python for Bioimage Processing & Analysis



- ▶ **Digital Images & Python:** what is a digital image? How do I deal with it in python?
- ▶ **Image Segmentation with Python:** what are semantic and instance segmentation? how do I perform segmentation on my fluorescence images? [Classical Methods, Ilastik (ML) & Cellpose (DL)]
- ▶ **Spot Detection:** how can we detect small spot-like objects such as fluorescent puncta, foci, beads, or RNA spots in fluorescence images? [Spotiflow (DL)]
- ▶ **Object Classification:** how can I classify objects in my images into different categories (e.g., mitotic vs. non-mitotic cells) to enable class-specific analysis? [Ilastik]
- ▶ **Measurements & Quantification with Python:** how can I extract quantitative data from my fluorescence images for plotting and drawing conclusions from my experiments?
- ▶ **Colocalization analysis with Python:** what is colocalization in fluorescence microscopy? How can I quantify it?





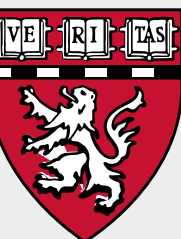
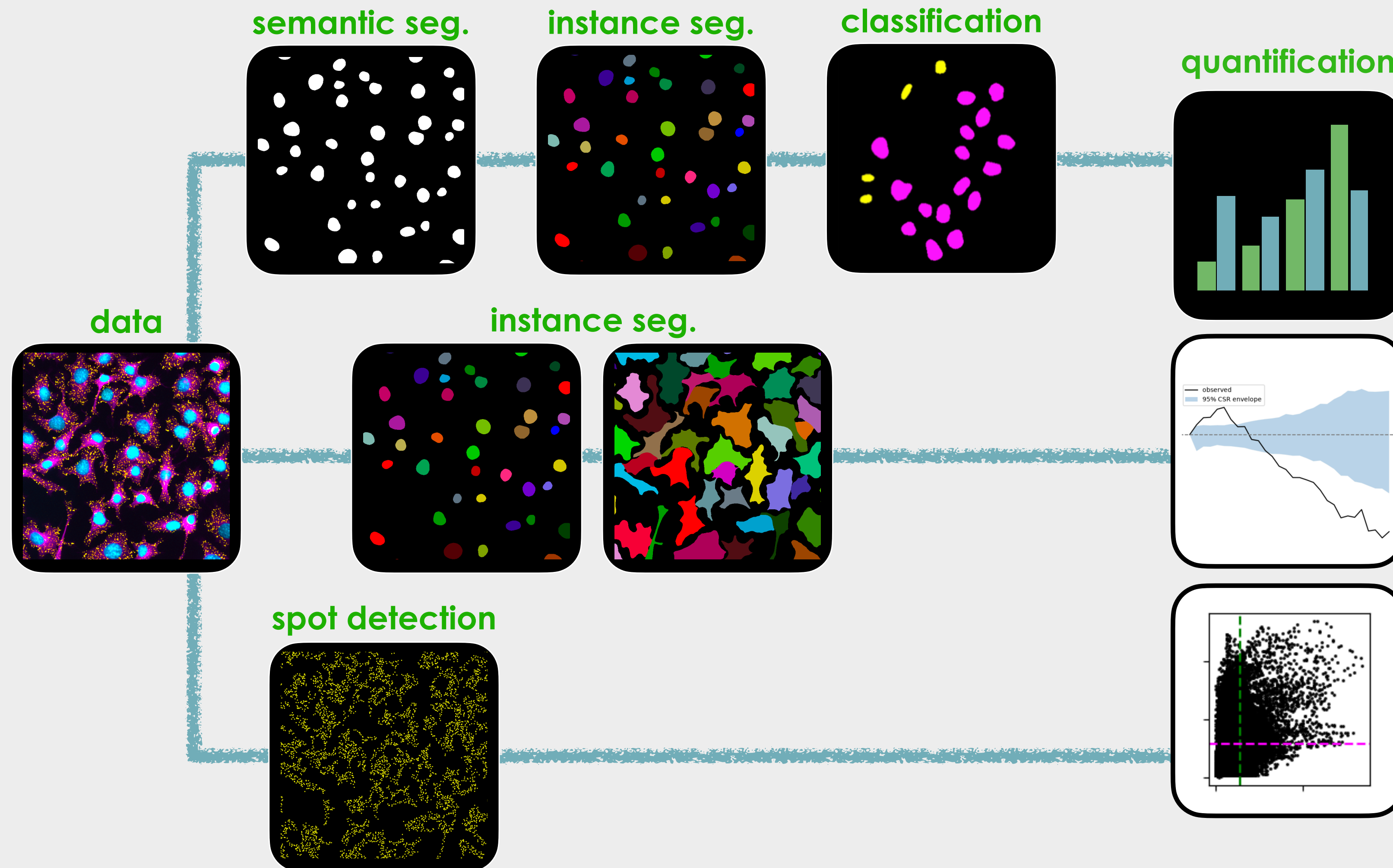
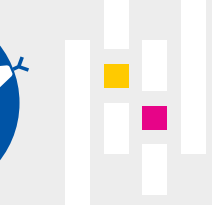
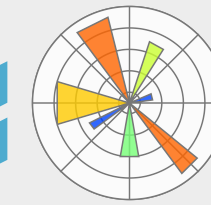
Course Topics II



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Learning Python for Bioimage Processing & Analysis

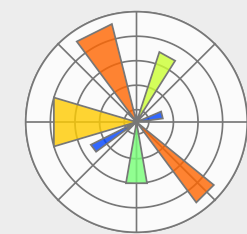
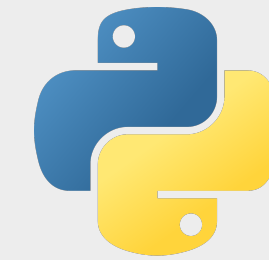


Learning Objectives

► ~~Become a Python & Bioimage Analysis Expert!~~



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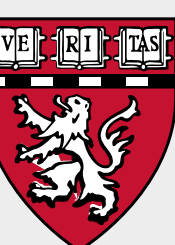
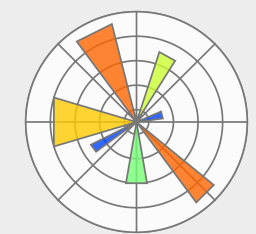
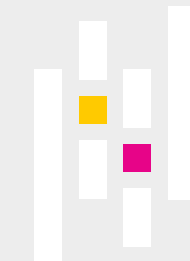
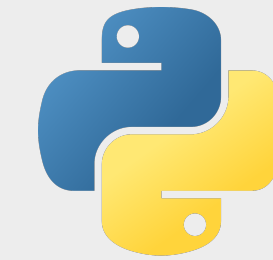
Learning Objectives

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- ▶ ~~Become a Python & Bioimage Analysis Expert!~~
- ▶ Gain a basic understanding of **what Python is**.
- ▶ Learn how to **get started with Python: installing Python**, setting up **virtual environments**, and launching **Jupyter Notebooks** and **.py** files.
- ▶ Get familiar with key **Python packages** commonly used in bioimage analysis.
- ▶ Learn how to **load, handle, and display images** using Python.
- ▶ Explore different approaches to **image segmentation** in Python, including classical methods, machine learning (Ilastik), and deep learning (Cellpose).
- ▶ Learn how to use deep learning for spot detection (Spotiflow).
- ▶ Understand the basics of **image quantification and measurements** and how to apply them using Python (e.g. single-cell measurements, spatial statistics, and colocalization).

(▶Improve LLM prompts)





Course Material



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Boston Bioimage Analysis Course | 2026

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Boston Bioimage Analysis Course | 2026

Welcome to the BoBiAC Book

Lecture Slides

Jupyter Notebooks

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Welcome to the BoBiAC Book

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02 - Python Basics

03 - Introduction to Digital Images

04 - Image Segmentation

05 - Spot Detection

06 - Object Classification

07 - Measurements & Quantification

08 - Colocalization

BoBiAC Exercises

BoBiAC Exercises

Course Materials Downloads

Downloads

Links

BoBiAC

uv

juv

pyrunner

Core for Imaging Technology & Education (CITE)

Past Editions

2025 Edition

Welcome to the BoBiAC Book

Welcome to the BoBiAC Book — your resource for the Boston BioImage Analysis Course (BoBiAC). This book is designed for **beginners** and provides a **hands-on introduction to image analysis using Python**. Inside, you'll find everything you need to follow the course: lecture slides, Jupyter notebooks, datasets, and step-by-step guidance through the material.

Lecture Slides

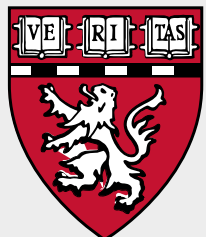
All [Lecture Slides](#) within the book are available for download as PDFs. You can download the complete slide decks from the [Course Materials Downloads](#) section of this book. Additionally, each individual page that contains lecture slides has a [Download the Slides](#) button at the top for convenient access to slides for that specific topic.

Jupyter Notebooks

All the [Jupyter Notebooks](#) within the book are available to download. You can download the complete list of notebooks from the [Course Materials Downloads](#) section of this book. Additionally, each individual page that contains Jupyter Notebooks has a [Download this Notebook](#) button at the top for convenient access to that specific notebook. All the notebooks can also be opened with [Google Colab](#) using the [Open in Colab](#) button at the top of each notebook page.



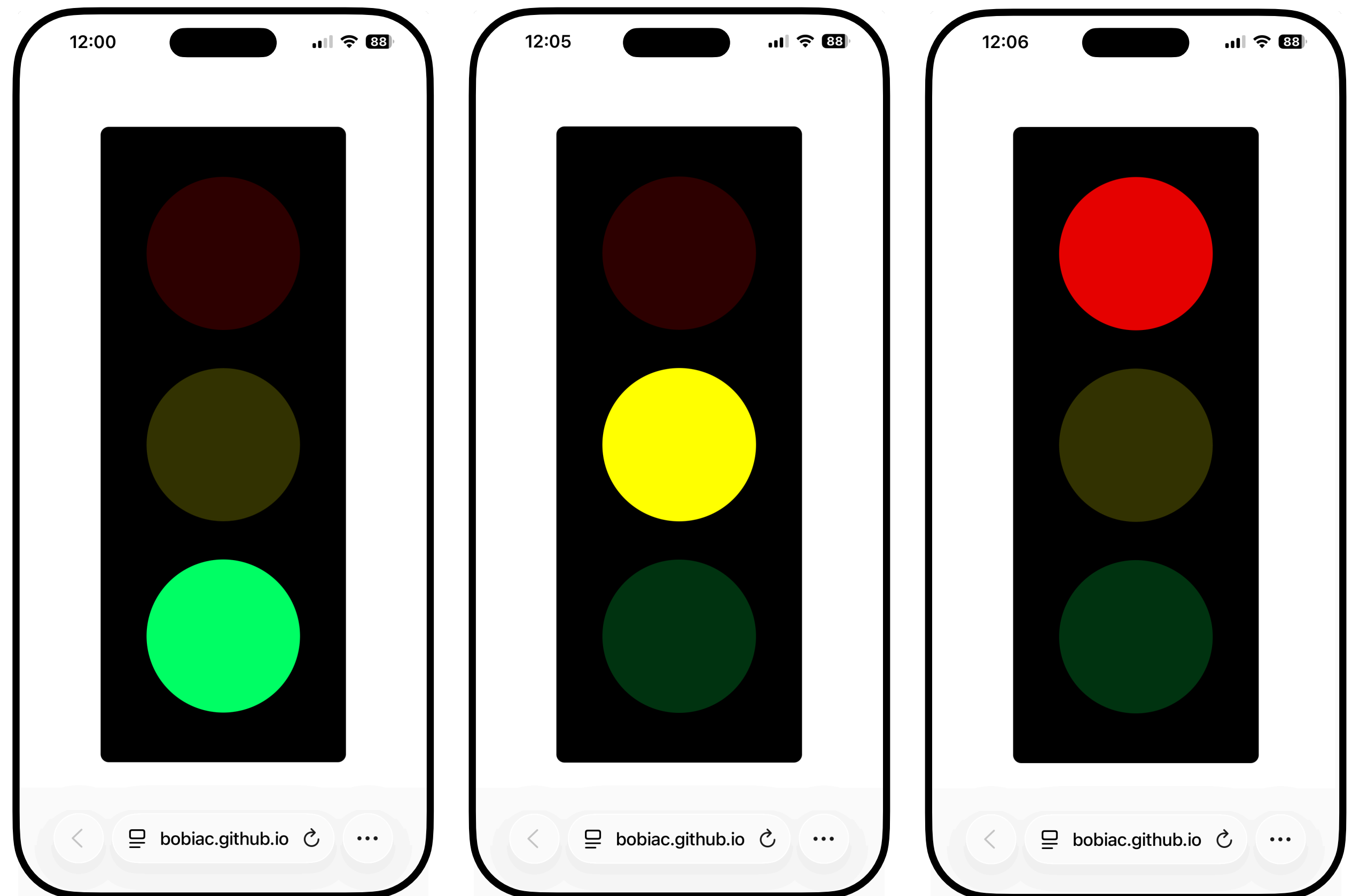
bobiac.github.io/bobiac-book





bobiac.github.io/classlight/bobiac2026

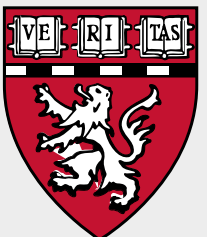
Realtime Feedback!



? Questions

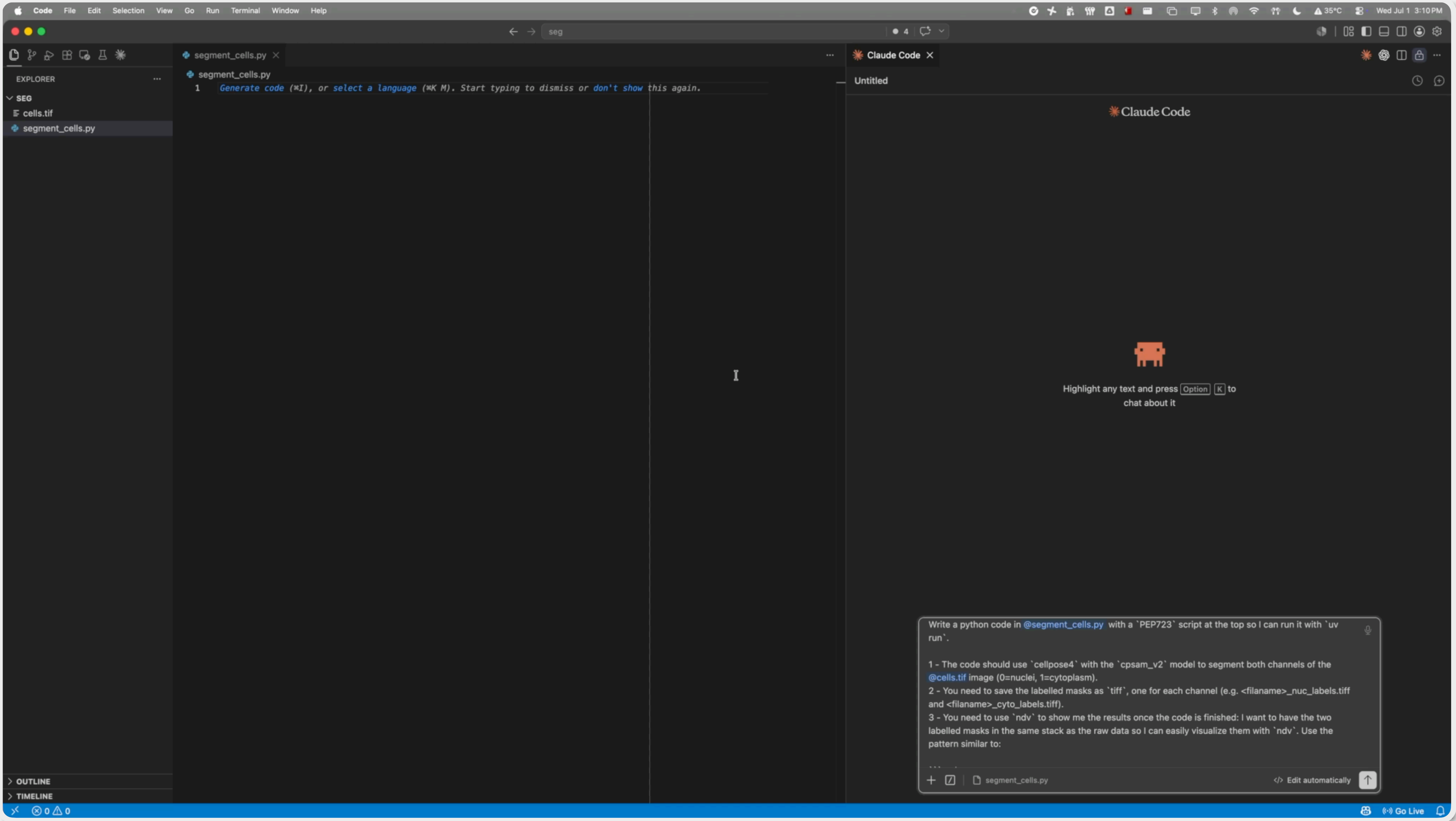
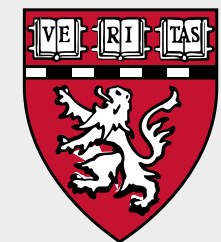


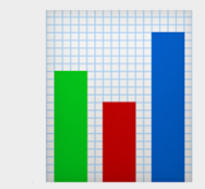
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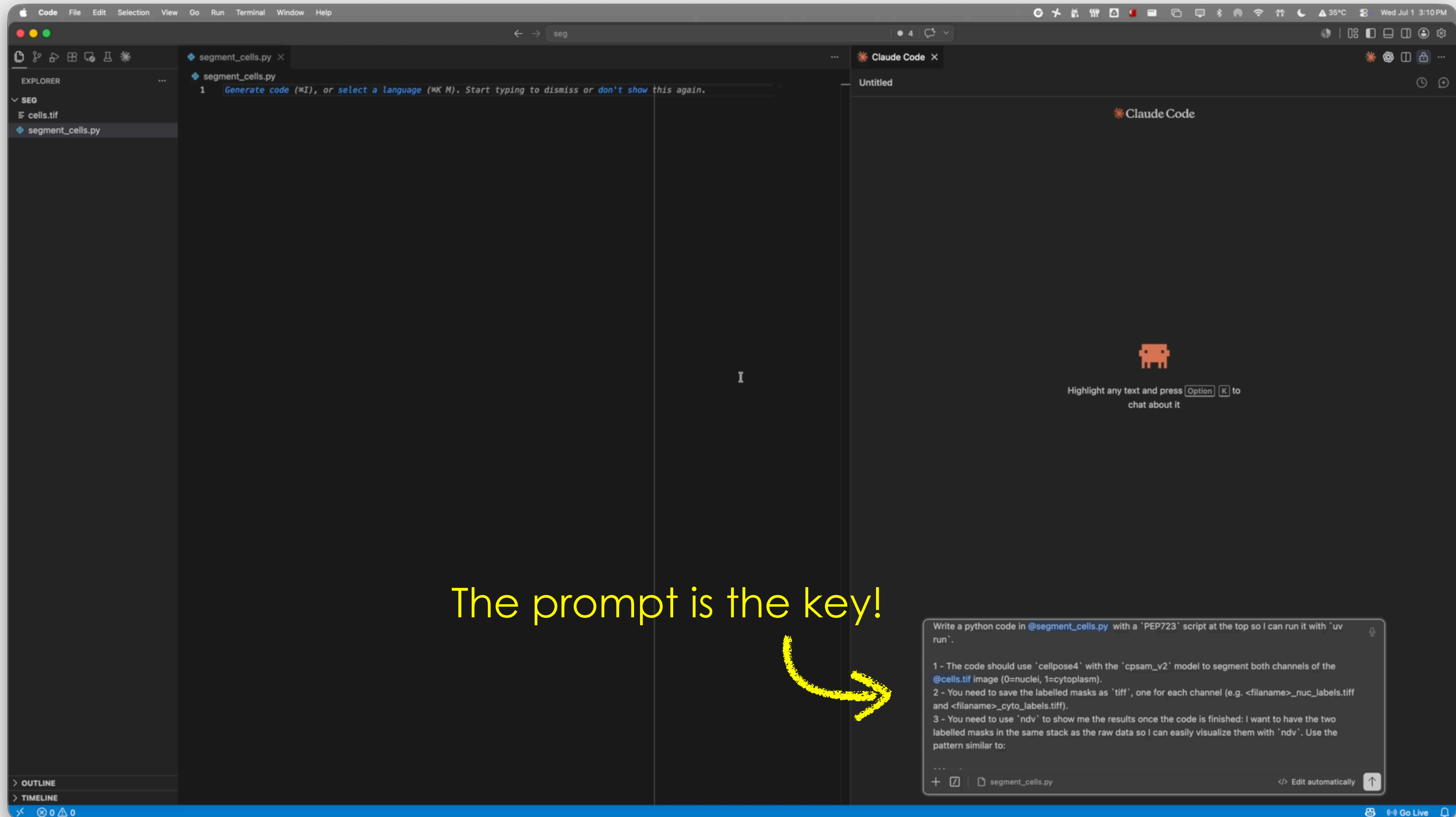


Why learn image analysis/Python if we have AI/LLMs?





Why learn image analysis/Python if we have AI/LLMs?



The prompt is the key!





1. My **name** is Federico
2. My **position** is as a Director
3. My **lab** is the CITE@HMS in Boston, MA
4. My model **system** is a microscope
5. I **acquired** my **data** with any microscope

